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Hand i2

Download STL files from the Gallery: i2RightHand (<https://inmoov.fr/inmoov-stl-parts-viewer/?bodyparts=i2RightHand>) i2LeftHand (<https://inmoov.fr/inmoov-stl-parts-viewer/?bodyparts=i2LeftHand>).

Before printing all the parts, you should print the CALIBRATOR (<https://inmoov.fr/wp-content/uploads/2019/01/Calibrator.stl>), to check if your parts will fit together. If you have a very hard time putting those parts together, adjusting the horizontal expansion setting of your slicer software can solve that, this setting can vary depending of your slicer and printer but users report to set it at -0.15 is a great place to start.

Here is the list of parts and the number of prints needed for 1 right hand:

- 1x i2_CoverFingerV3
- 1x i2_FingersMoldX5V3
- 1x i2_FingersTipX5V2
- 1x i2_FingersX5V1
- 1x i2_HandCoverV1
- 1x i2_PalmCoverV2
- 1x i2_PulleyX5V1
- 1x i2_WristGearV1
- 1x i2_WristLargeV2

Print WristLarge, FingersMold, Fingers, WristGear, Pulley, resolution 0.25 maximum with an infill of 30%, wall thickness 2mm, no support, no raft.

Print CoverFinger, FingersTip, HandCover, PalmCover, resolution 0.25 maximum with an infill of 30%, wall thickness 2mm, support required, raft if your printer requires it.

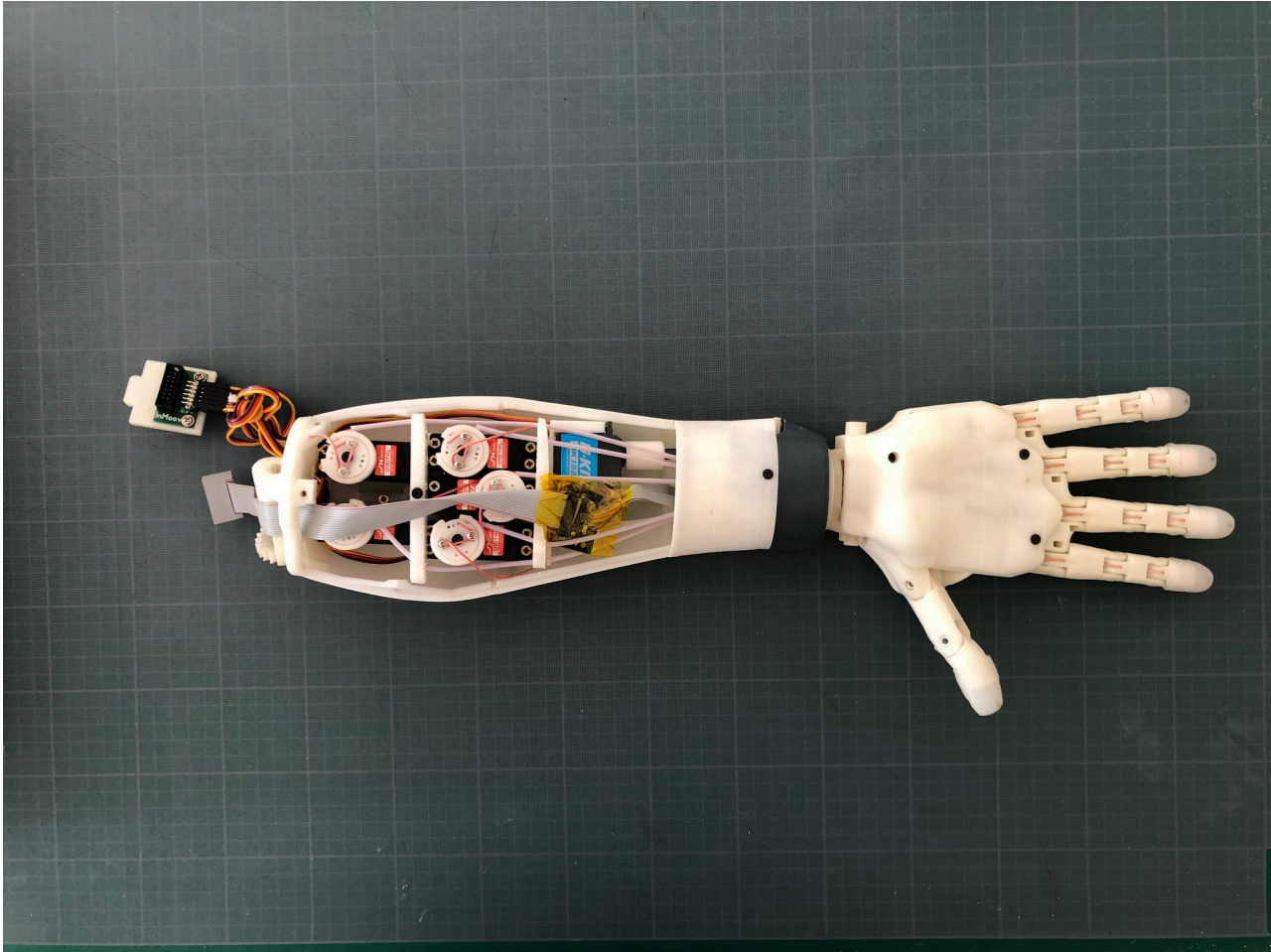
BOM:

- 25x countersunk head screws M3 x 16mm
- 5x countersunk head screws M3 x 4mm
- 1x countersunk head screws M3 x 12mm
- 5x flat head screws M1.5 x 4mm



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- 5x Hall sensor AH3503
- 5x disc Magnet diameter 2.5mmx1mm
- 1x teflon pipes ID1.5MM X OD2.5MM, 3 meters

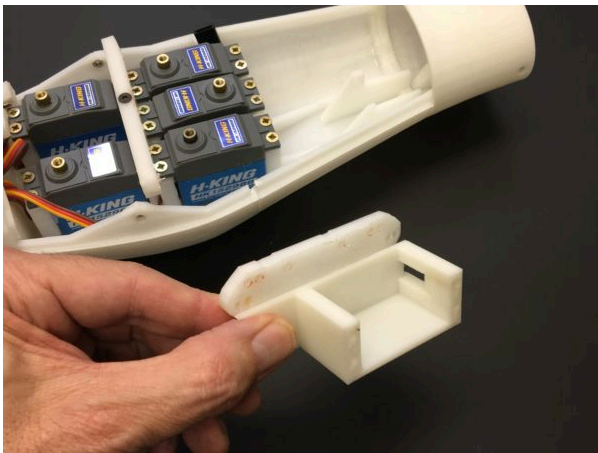




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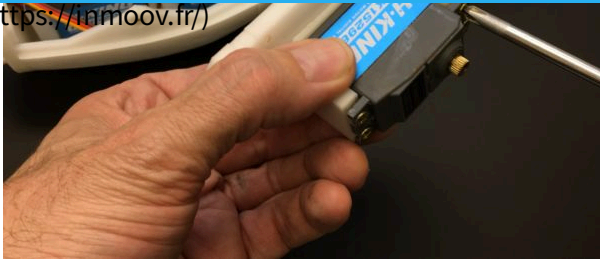
Step1



Glue the wrist servo support to RobCableFront



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Mount the servo motor with 4 screws



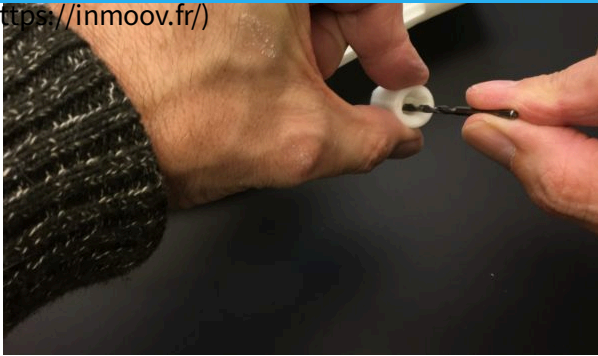
Redrill holes if necessary with a M3 drill bit



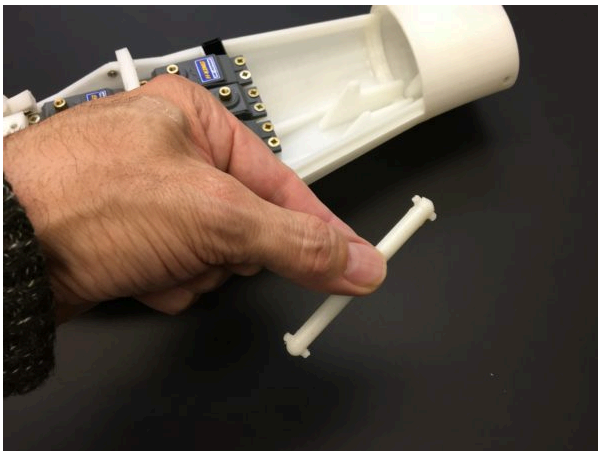
Redrill holes to allow the M3 PTFE tubes to run through them



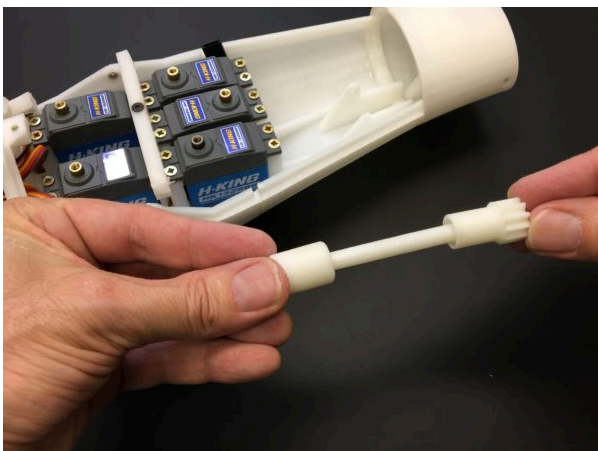
(<https://inmoov.fr/>)



Redrill hole with a M3 drill bit

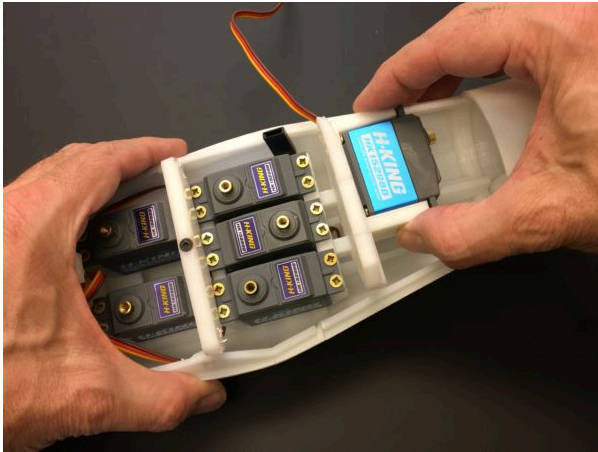


Glue together the two shaft printed parts

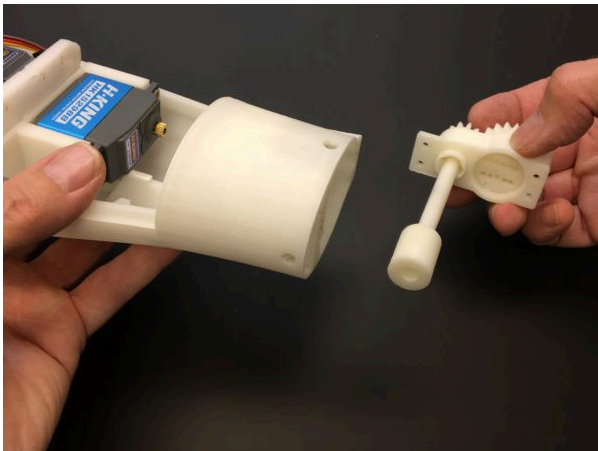




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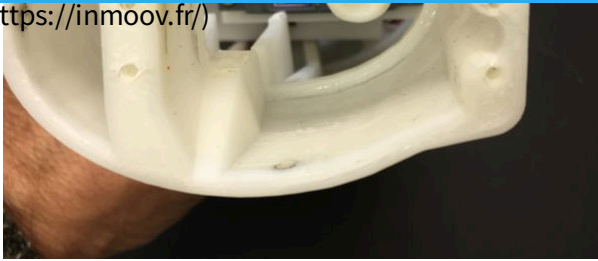
Mount RobCableFront and the servo support to the servo bed



This view is to show you the assembly of WristGear and how those parts go together



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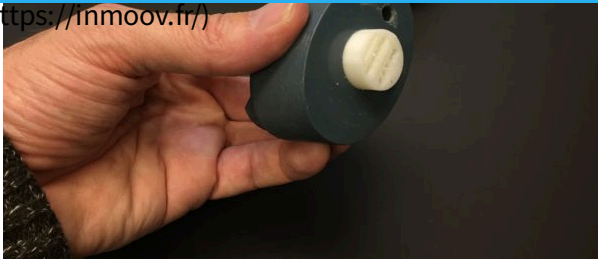
Set your servo approximately at 90 degrees(HK15298B) or 150 degrees(JX-6225MG) . (Midway of it's total range). You can use this script (<http://www.inmoov.fr/wp-content/uploads/2013/10/Arduino-Schetch-servo-zeroestmax.txt>) to do that. Mount with a 3mm screw the end tip of the assembly to the servo shaft



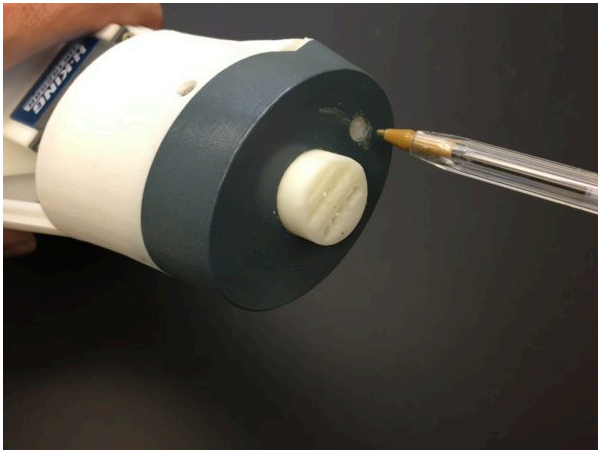
Screw the assembly to the wrist with 4 screws



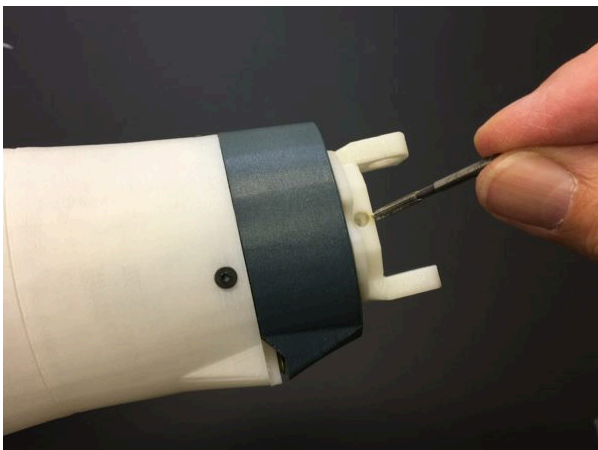
(<https://inmoov.fr/>)



Assemble the main gear as shown



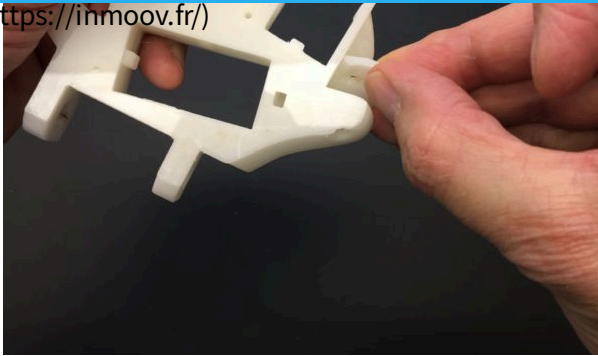
Note the head of the small gear has to be fitting in the reserved hole of RotaWrist2



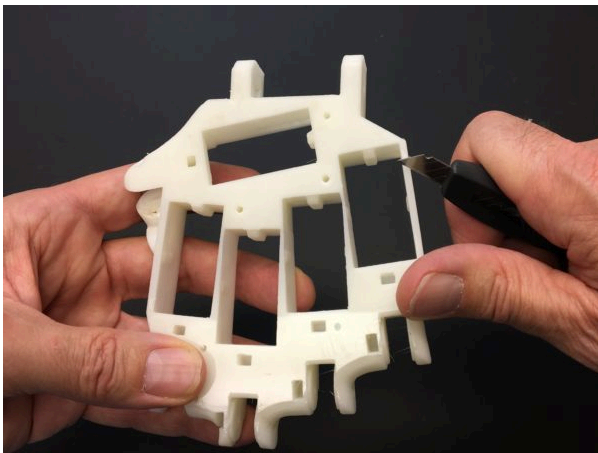
With a M3 tap, align the main gear and RotaWrist3, and add a M3 x 12 mm screw to fix them together



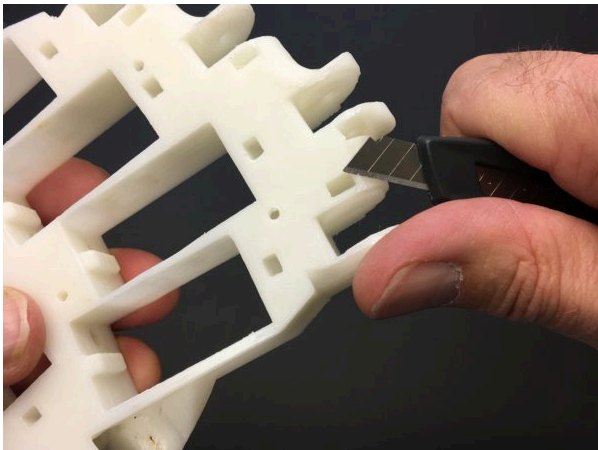
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Remove that piece of printed pre-support, You don't need that anymore



With a sharp blade, clean up the edges if necessary

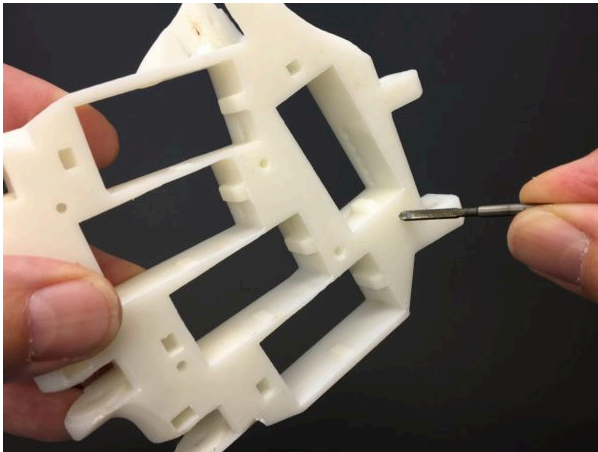




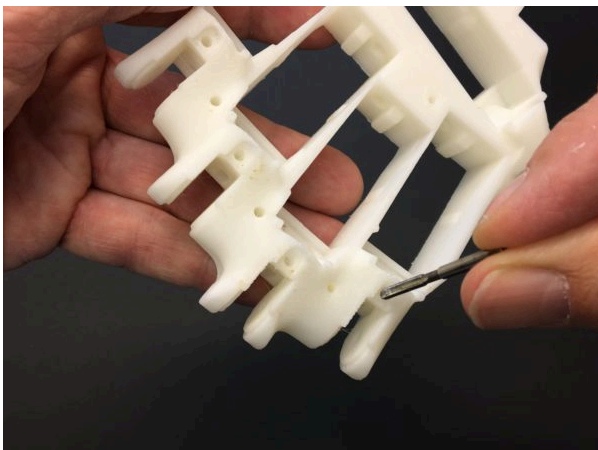
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Redrill with a M8 drill bit WristLarge

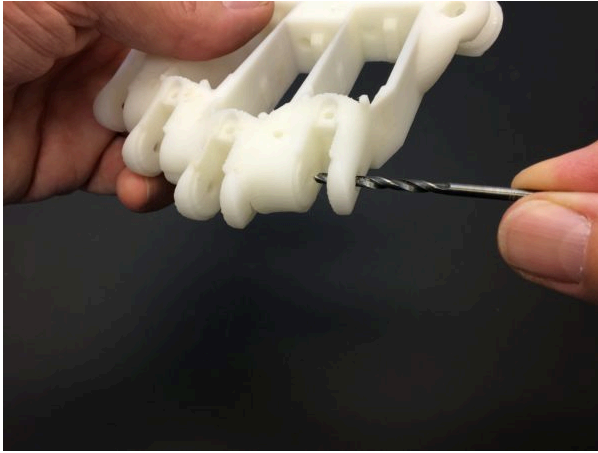


With a M3 tap, prepare the holes for some M3 x 16 mm screws where the hand covers are going to be fixed

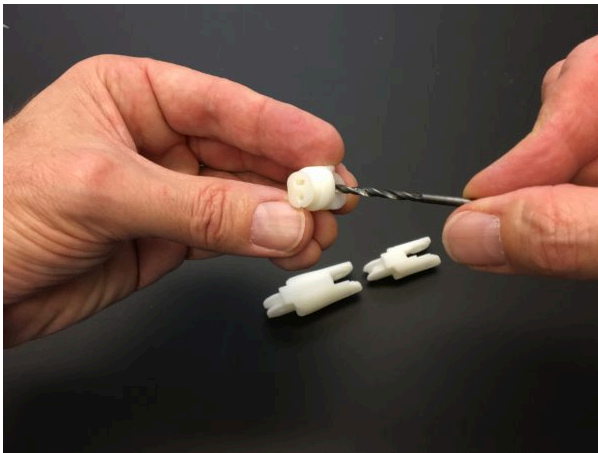




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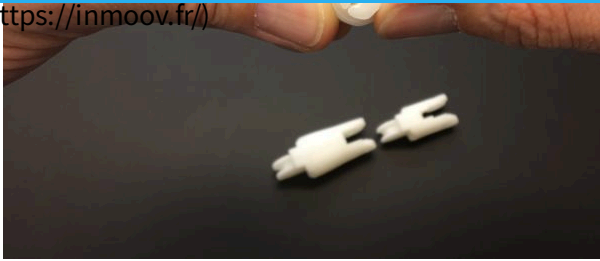
With a M3 drill bit, clear up all the holes which will attach each finger



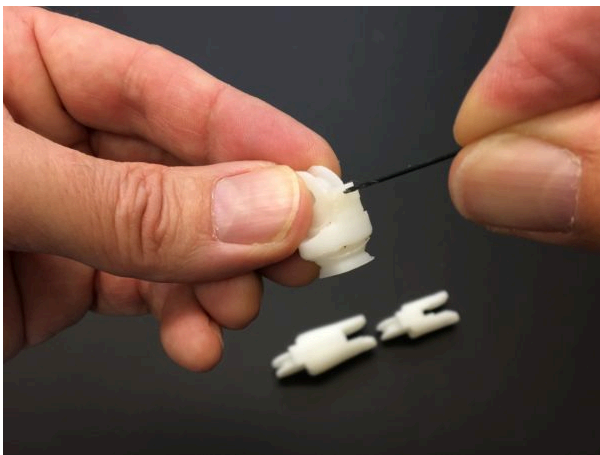
With a M3 drill bit, clear up all the holes of each 3 sections of all fingers



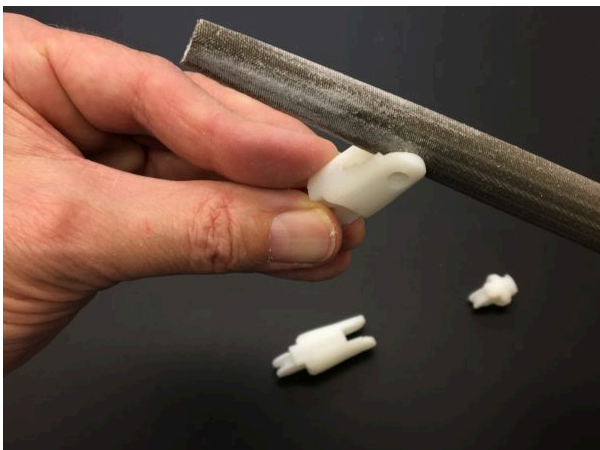
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With a sharp blade, clear up the exit for the hall sensor

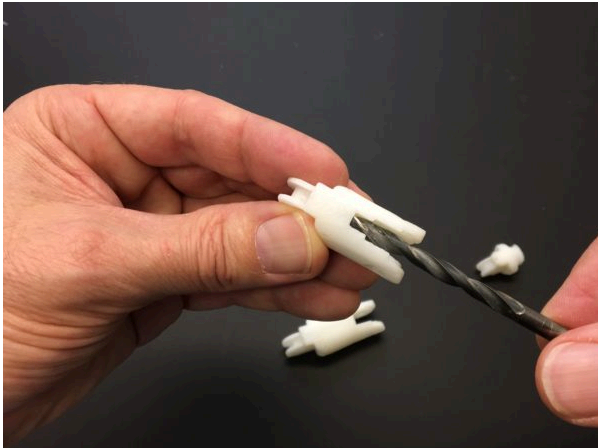
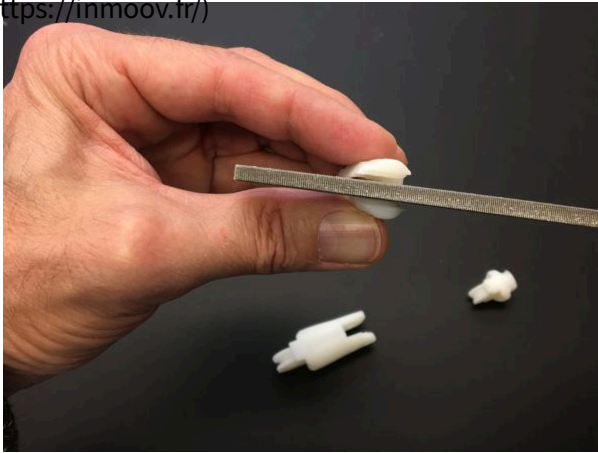


With a M1.5 drill bit, clear up the hole where the end of the spring will be attached





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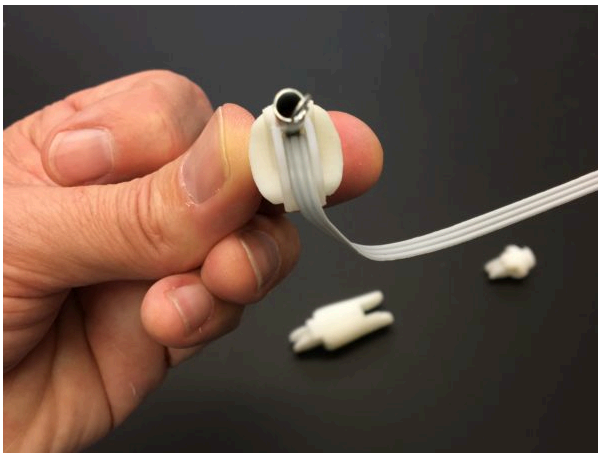
With a M5.5 drill bit, clear up this finger section to allow the spring to run through properly



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With a M5.5 drill bit, clear up this finger section to allow the spring to run through properly



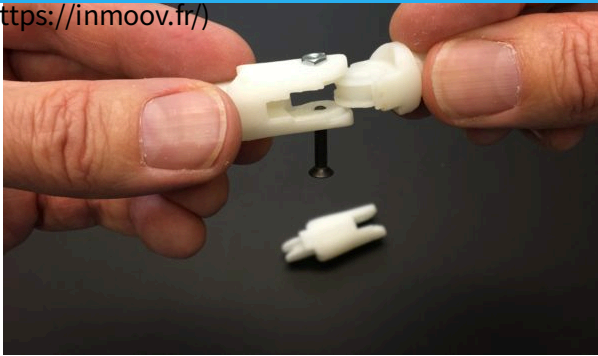
This is to show you how the spring and the three sensor wires will be mounted in each finger section



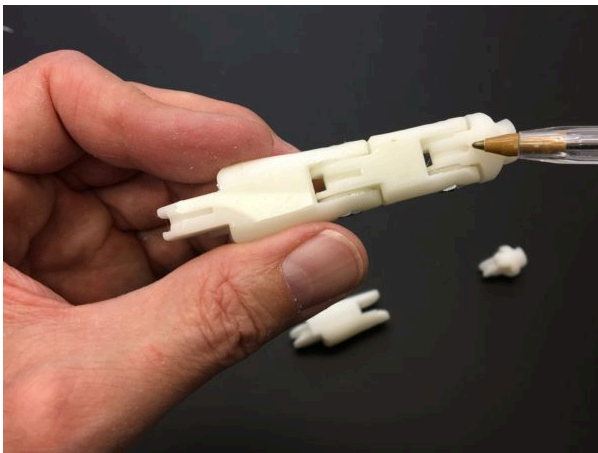
Make sure a needle can slide through each finger section as shown, otherwise clear up the holes



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Add and mount finger sections together using M3 x 16 mm screws and bolts



Make sure the entrances for the braided fishing line are cleared up as well

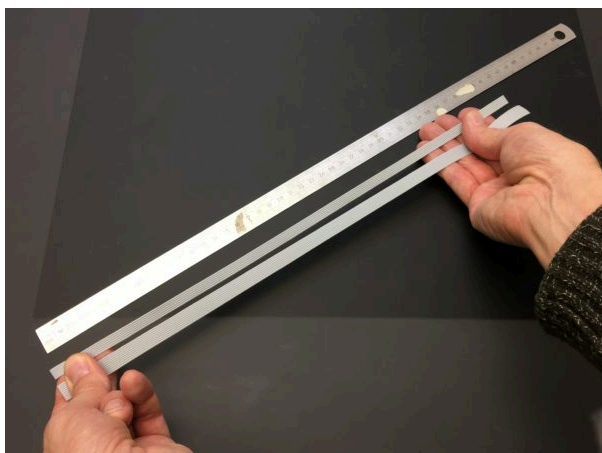




(<https://inmoov.fr/>)



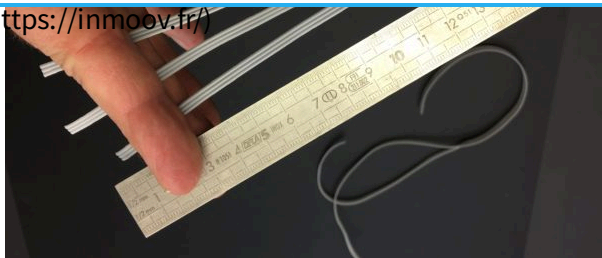
Prepare a length of ribbon cable of 40cm. You can google search with these keywords: ribbon cable 16 pin (make sure to get the most flexible and thin as possible)



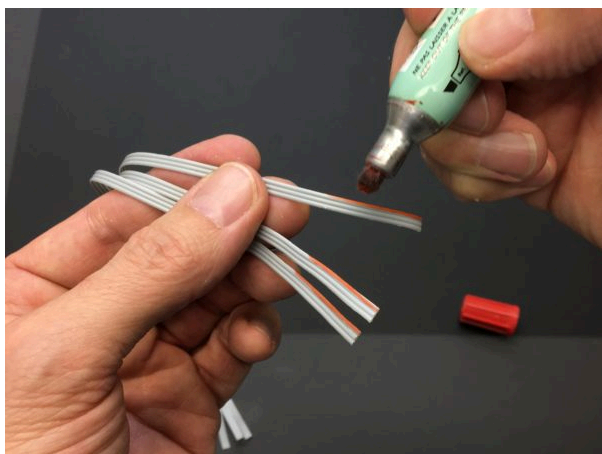
Separate the ribbon to have a length of 40cm with 6 wires and another one with 10 wires



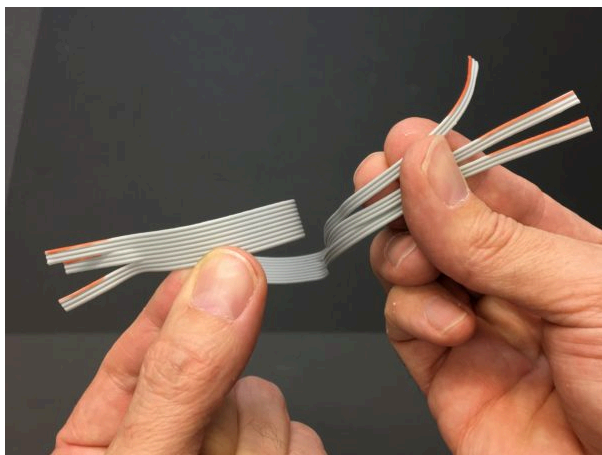
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*Remove completely 1 wire from the 10 wires section.
Separate the 9 wires section to have three sections of
3 wires on a 15cm length*



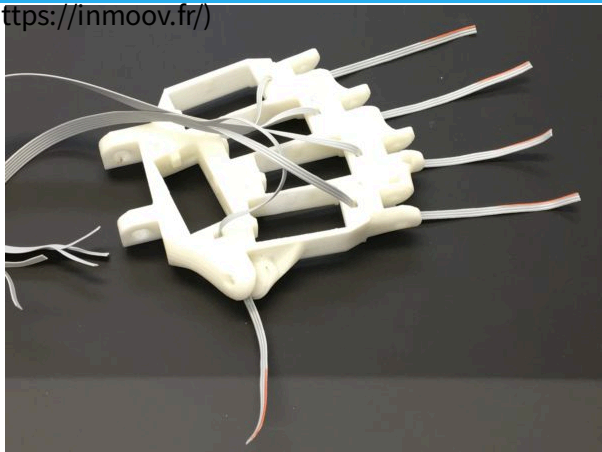
*With a red marker, color one wire on each section. We
will assume these are the positive (+) wires*



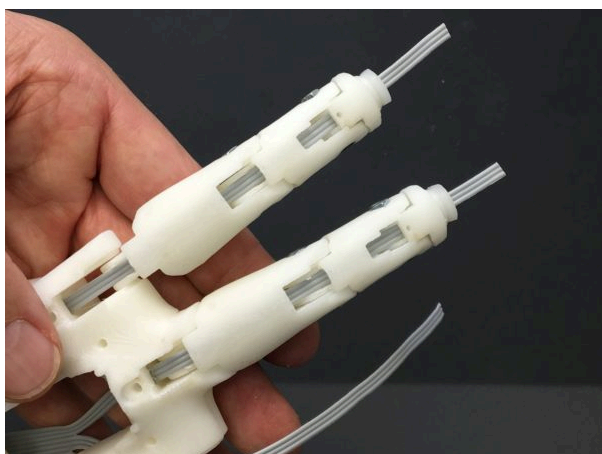
*Do these coloring on both end of the ribbons, this will
be very useful to avoid mixing wires when we will do*



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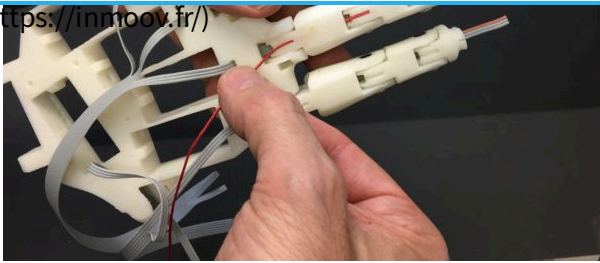
Run through WristLarge each 5 ribbon sections



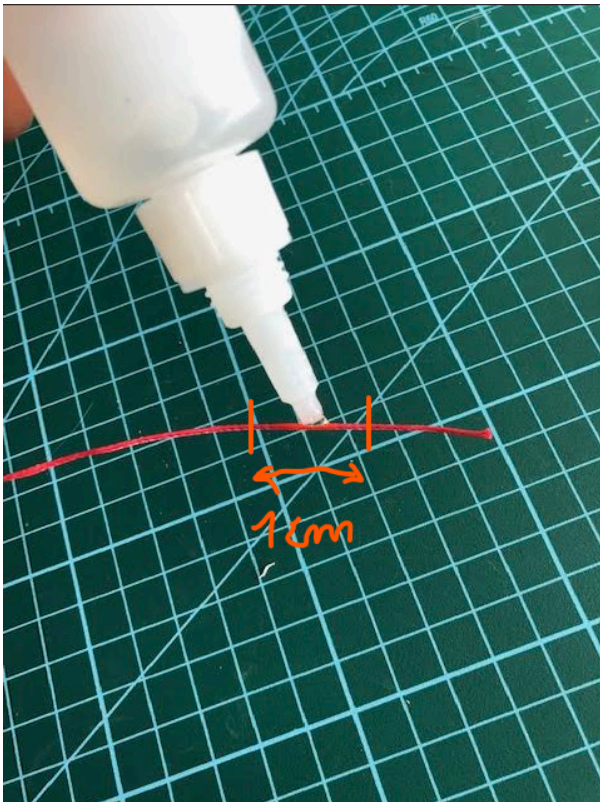
Run the ribbons through each fingers, look out not to mount the fingers upside down. Also note that each finger are similar, only the thumb differs with it's shape.



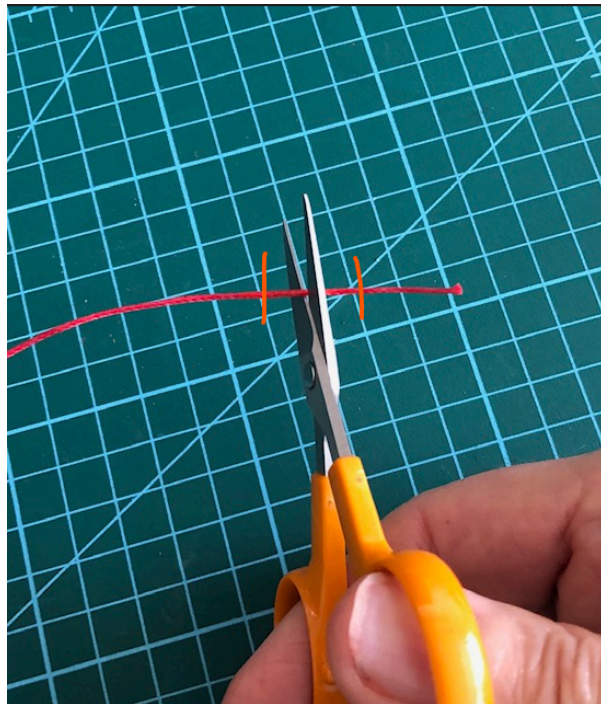
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Now run the braided fishing lines through each fingers.
To make the fishing line stiff on the end tip, it is useful
to add some superglue.



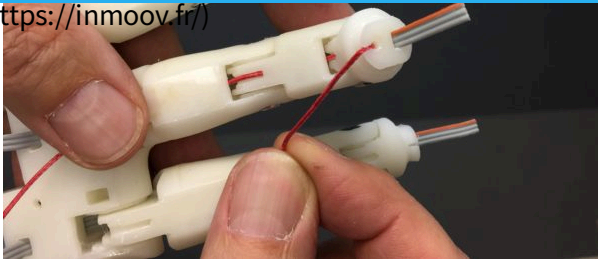
Add superglue on a 1 cm length of the braided fishing
line.



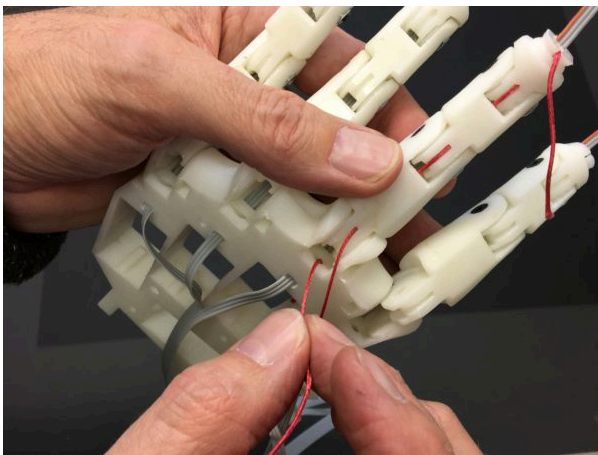
Wait until it is completely dry and cut the glued section
in the middle, this will ensure a strong and thin tip.



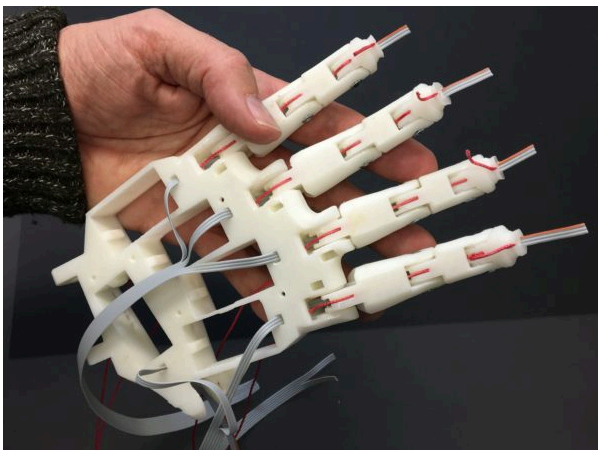
(<https://inmoov.fr/>)



By pulling the fishing line as shown into the V groove of the finger tip, you can get attached

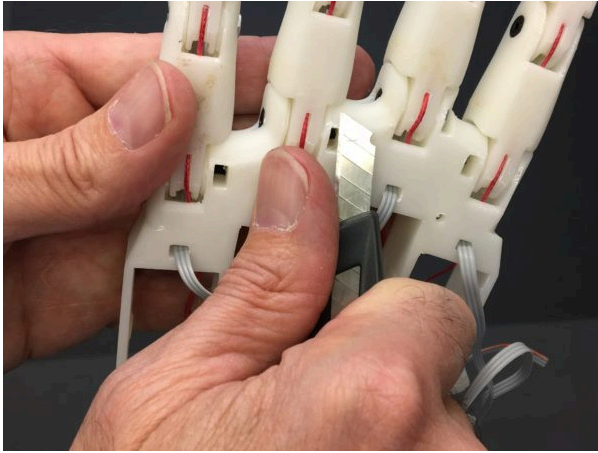


Now run the other tip of the fishing line into the hole of WristLarge

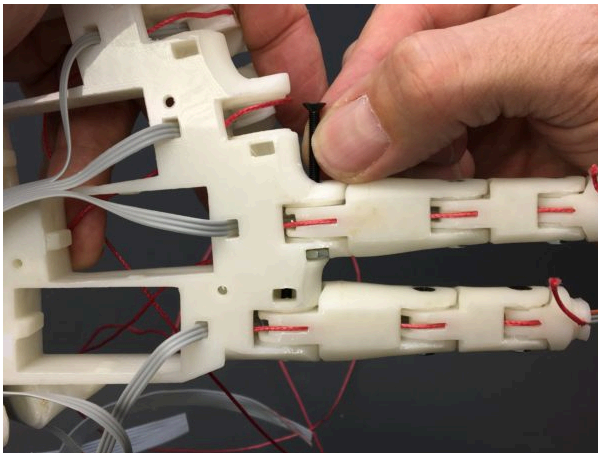




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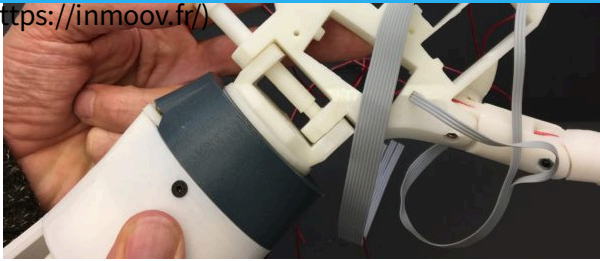
If you haven't done it, clear up the 3mm bolt casings for the next step



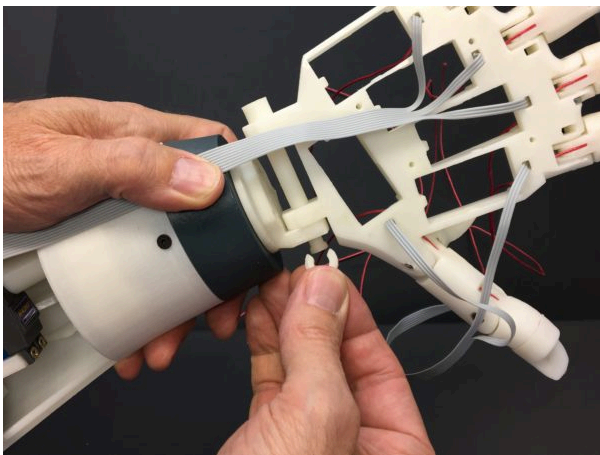
It's now time to add the M3 x 16 mm screws and their respective bolts to assemble the fingers to WristLarge. Start with Majeure finger, then Index finger, Ring finger, Pinky and finish with Thumb



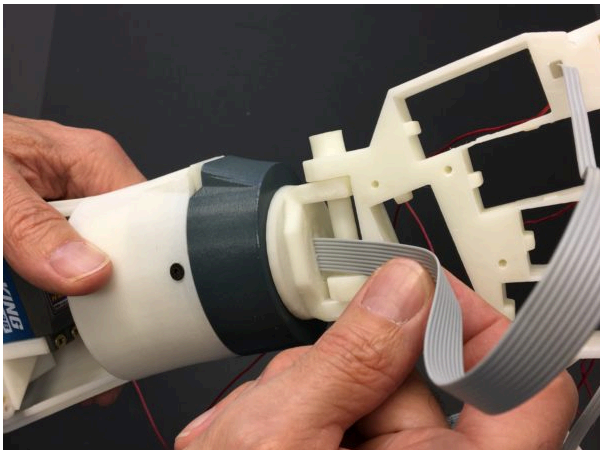
(<https://inmoov.fr/>)



Mount WristLarge to RotaWrist3 with the appropriate M8 printed bolt



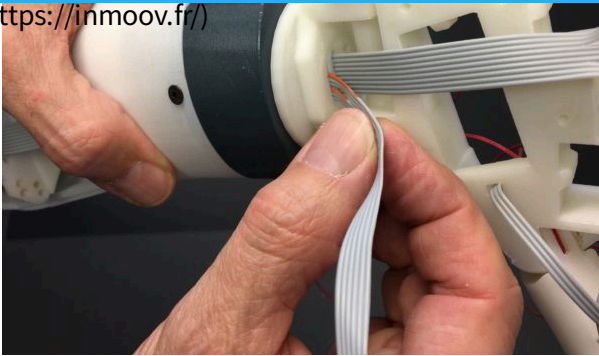
Set the mini clamp on the groove of the printed bolt



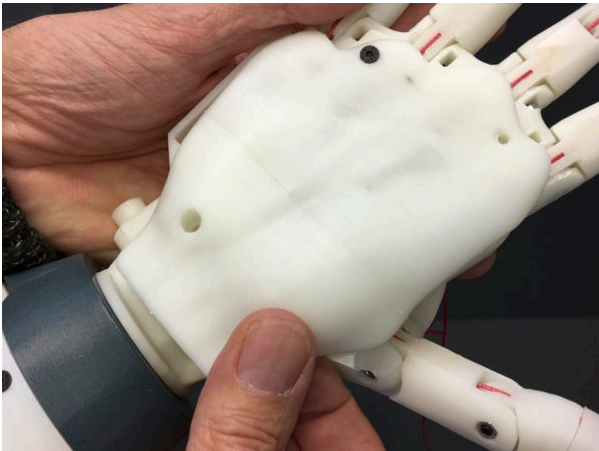
Run through the main gear the largest ribbon



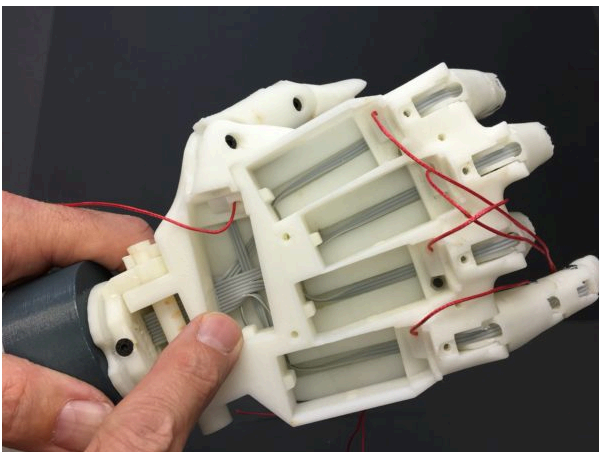
(<https://inmoov.fr/>)

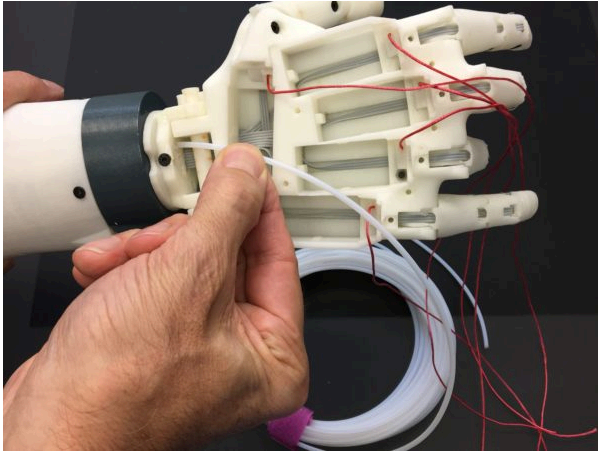


Then run through the main gear the second ribbon

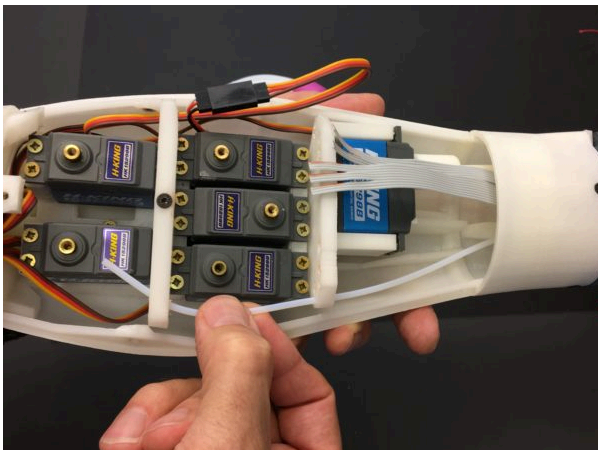


Place and screw the palm cover with 3 screws of M3 x 16 mm



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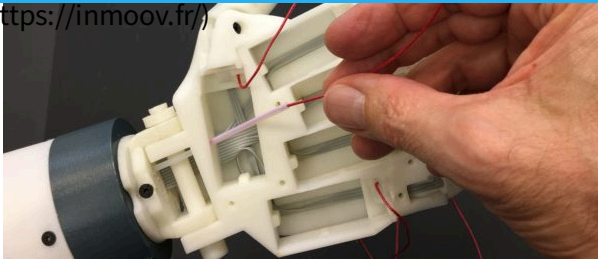
Now run the M2.5 PTFE tubes through the main gear
Teflon pipes ID1.5MM X OD2.5MM
(<https://fr.aliexpress.com/item/32730855848.html?spm=a2g0s.9042311.0.0.3b966c37jungcW>) The PTFE tubes are with the following lengths: Thumb 335mm
Index 300mm Middle 300mm Ring 300mm Pinky 360mm



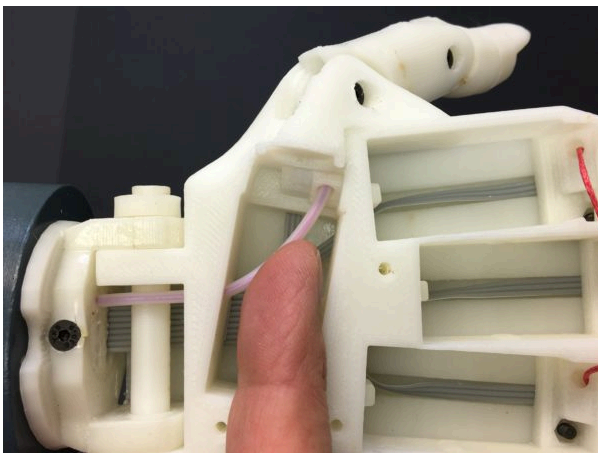
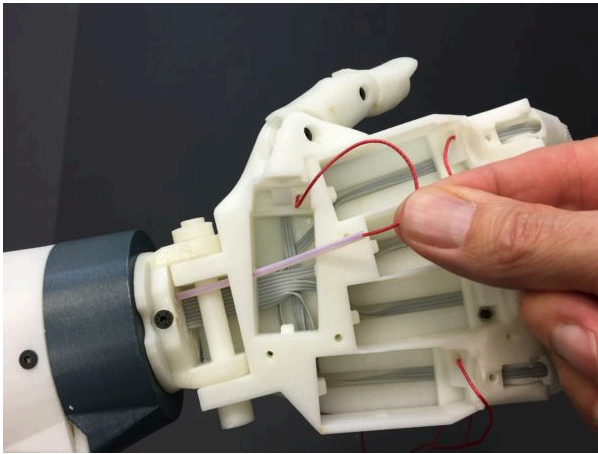
Run the tube through the the servo bed holes.
Approach the tube to the servo shaft at about 1.5cm



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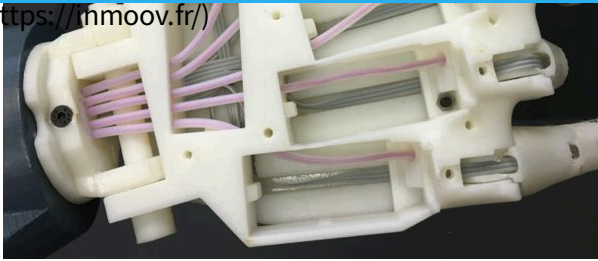
Run the fishing line through the tube



*And place the end of the tube into the reserved hole
until it cannot go further*



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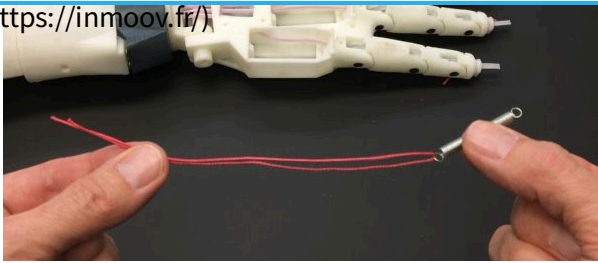
Repeat the operation for each finger



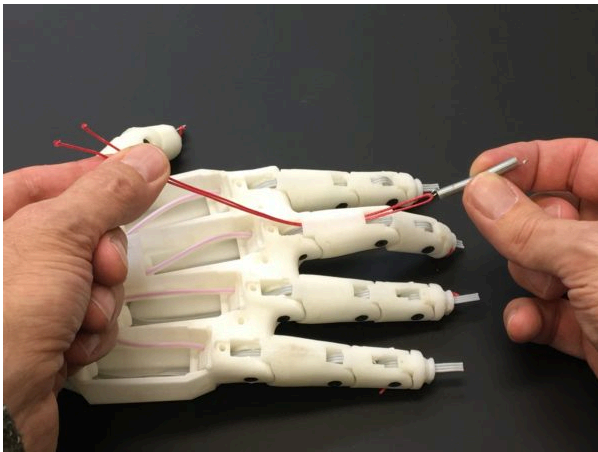
Now get your 5 springs Springs 3/16"x1-3/4
(<https://www.ebay.com/itm/200pc-Spring-Assortment-Set-Compression-Carburetor-Extension-Z1T1/264399686883?epid=19011685359&hash=item3d8f730ce3:g:BGUAAOSwRrJdLUQ5>)



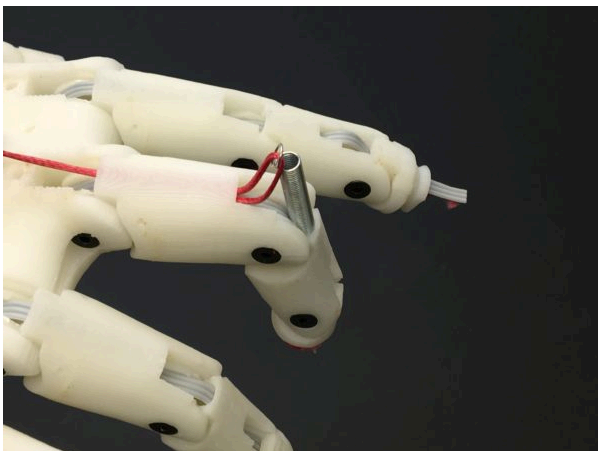
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Run through one side of the spring a piece of fishing line. The total length of the fishing line should be about 30cm

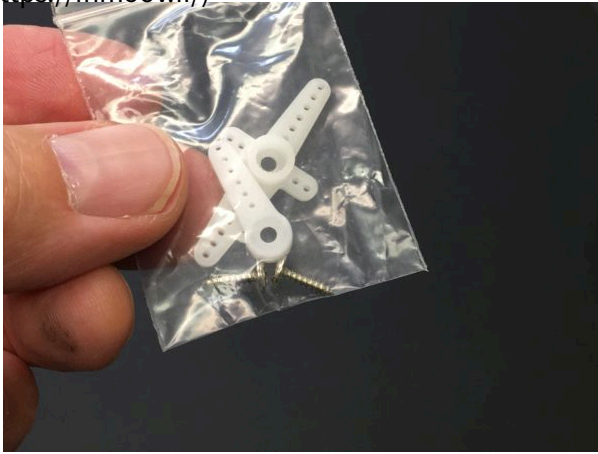


Run the fishing line through the spring hole of the main finger section

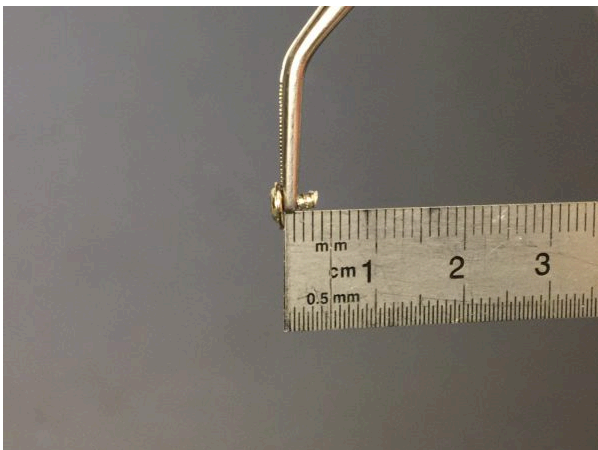




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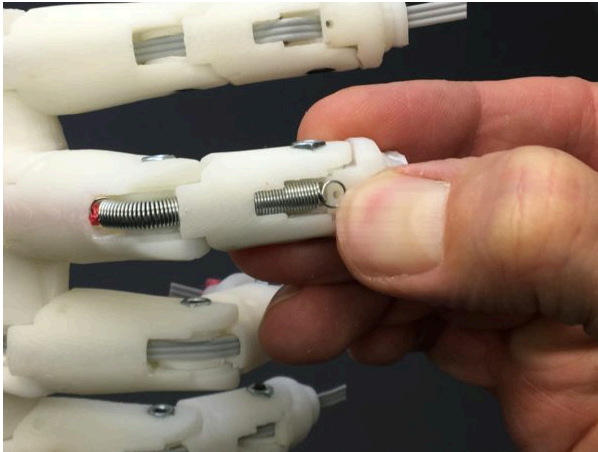


Get 5 small screws, I used some from the servos that are mounted in the eye mechanism, the diameter of these screws is about 2mm

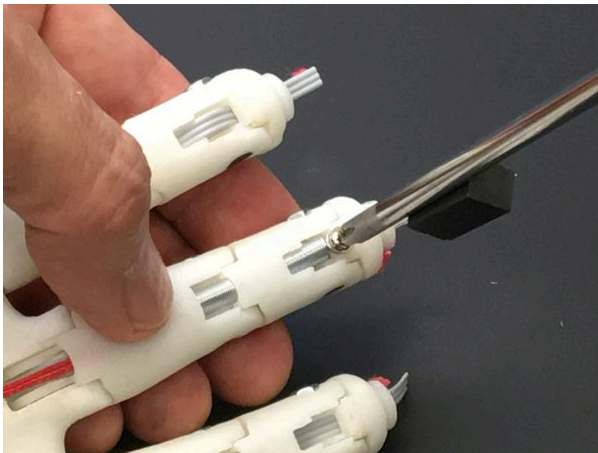




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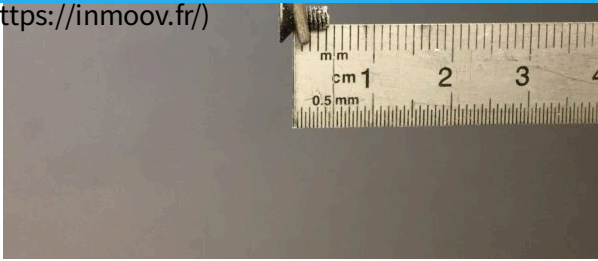
Hold the spring in place



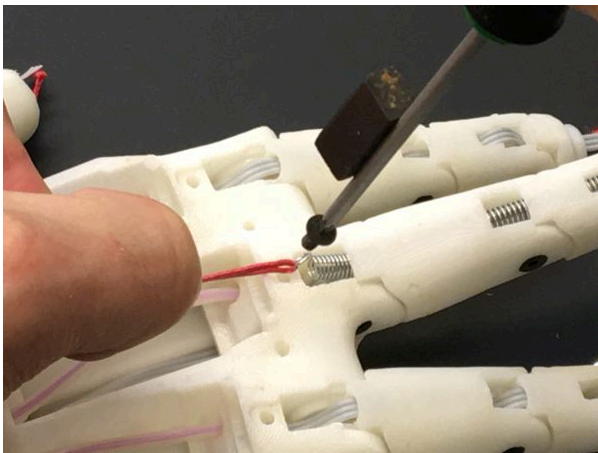
*And mount that small screw to set the spring in place.
Make sure the screw is not piercing the three ribbon
wires*



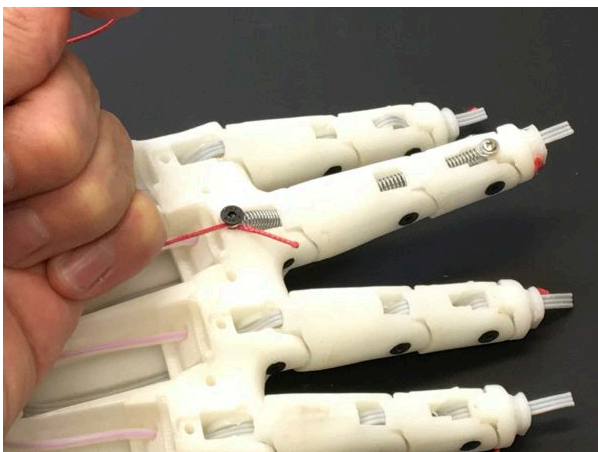
(<https://inmoov.fr/>)



Get or cut five screws of 3mm with a maximum length of 4mm



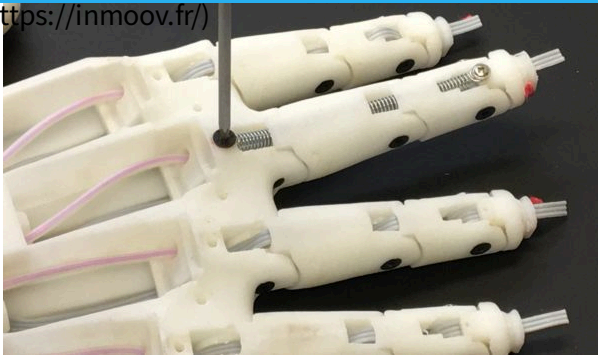
Pull the fishing line to get the spring tip to be in place and mount the screw. Note, I use a black magnet on my screw driver to ease the action of setting the screw



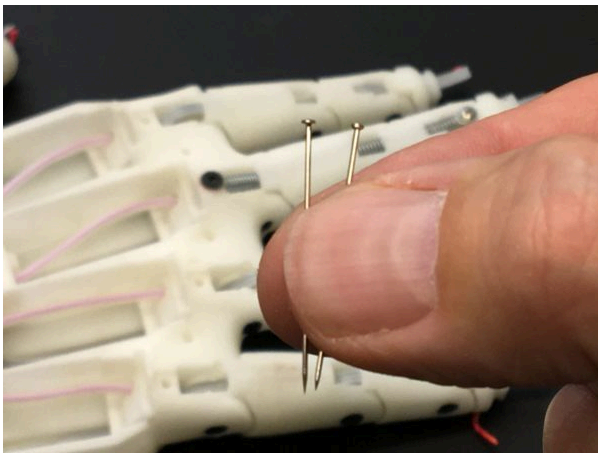
Before tightening the screw, withdraw the fishing line



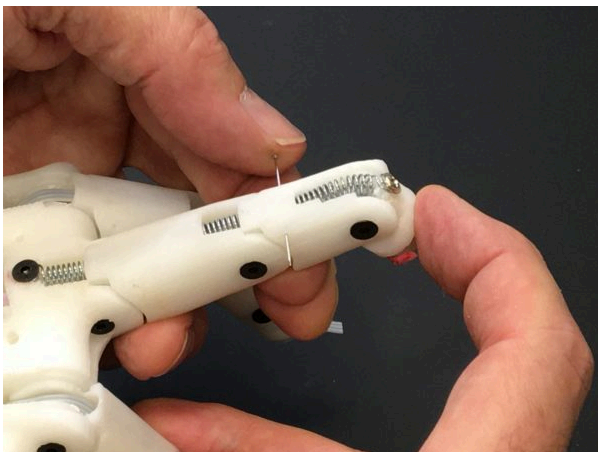
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Finish tightening. Make sure you are not piercing the three ribbon wires

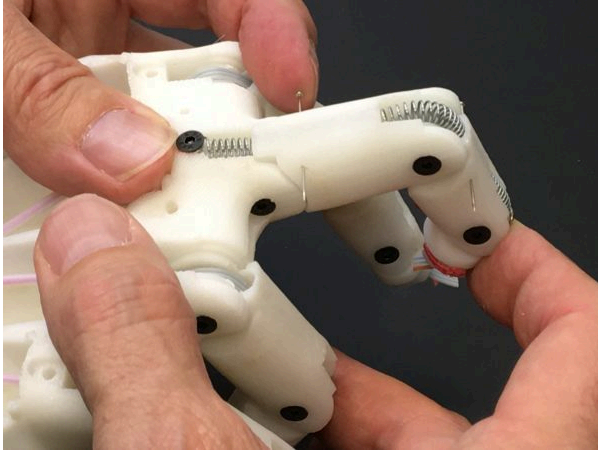


Get two needles

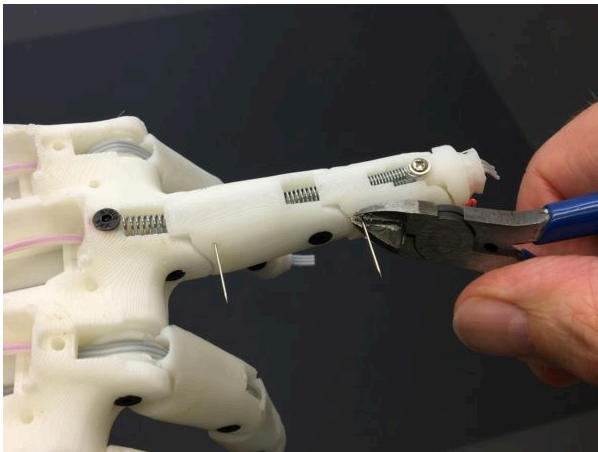




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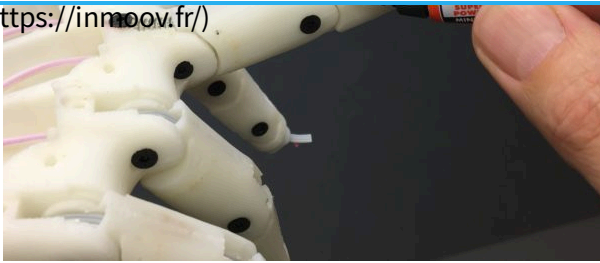
Fold the second section of the finger and run the second needle through the finger and the spring until it goes through. This allows to set adapted tension on each finger section



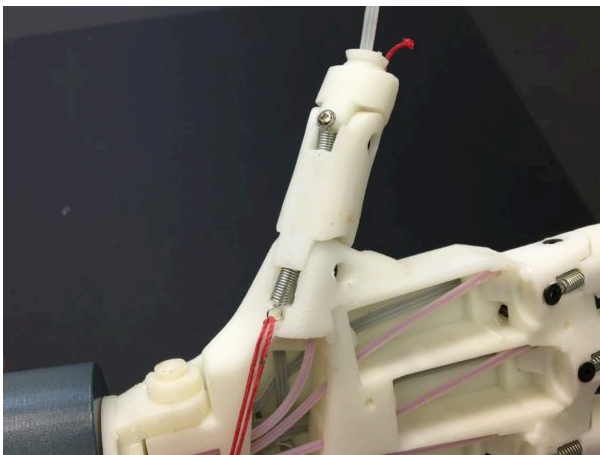
Cut the needles with a small pliers



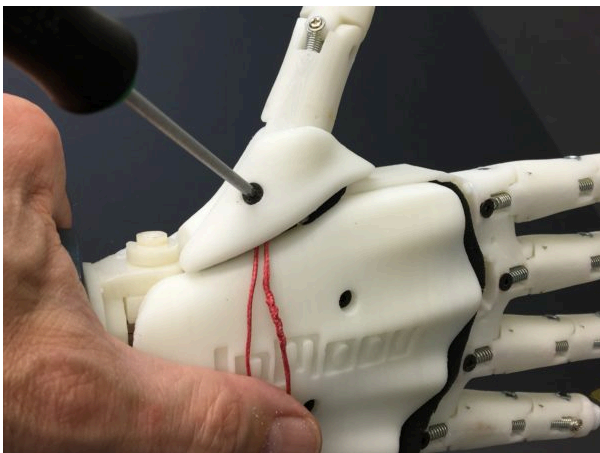
(<https://inmoov.fr/>)



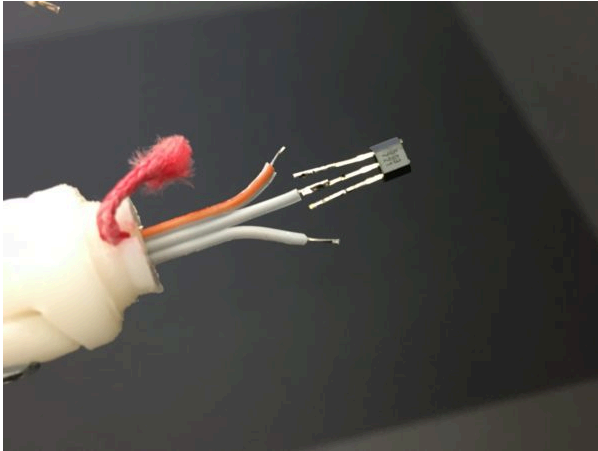
Put a drop of super glue to ensure the needle won't come out anymore



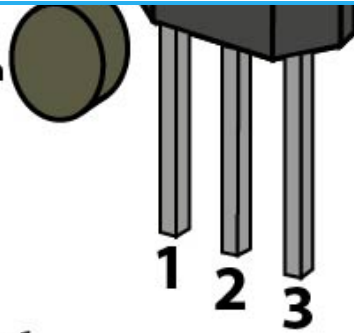
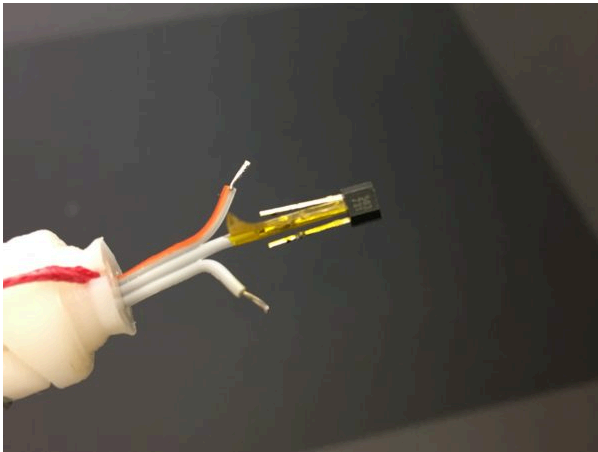
Time to do the same with the thumb, there is only two finger sections, so the spring is less in tension and doesn't require needles



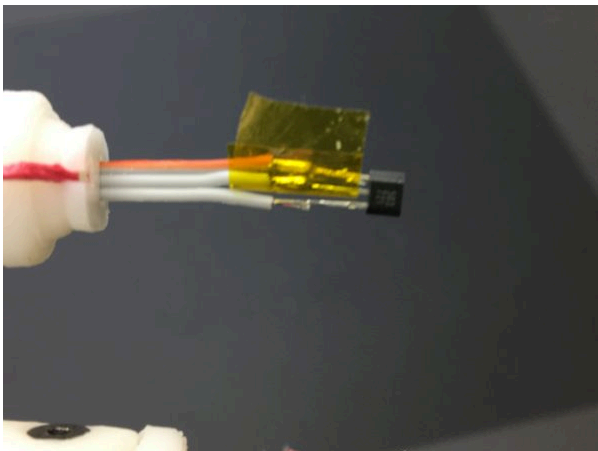
(<https://inmoov.fr/>) you still need to access the inside



Let's solder the hall sensor to the wires. Start soldering the middle wire. Note the position of the sensor on my picture, because one side is positive and the other is negative. HALL sensors AH3503 (https://fr.aliexpress.com/item/32899671134.html?src=google&src=google&albch=shopping&acnt=494-037-6276&isdl=y&slnk=&plac=&mtctp=&albbt=Google_7_s hopping&aff_platform=google&aff_short_key=UneMJZVf&&albagn=888888&albcpr=6459793138&albag=77316928277&trgt=743612850714&crea=fr32899671134&netw=u&device=c&gclid=CjwKCAiA_f3uBRAmEiwAzPu aM3NMIs2FXydv7C1B073Ng0vvuwXhDTT_k_XjAtt_- IML73q4LzrOxoCUzkQAvD_BwE&gclsrc=aw.ds)

[\(https://inmoov.fr/\)](https://inmoov.fr/)**Magnet****Diameter 2x2mm****1=+VCC****2=GND****3=Output**

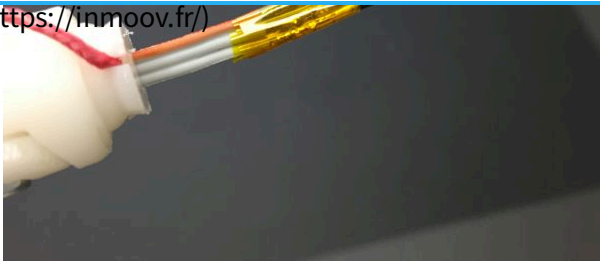
Protect your solder with a small piece of tape. I used Kapton tape. This is delicate to do.



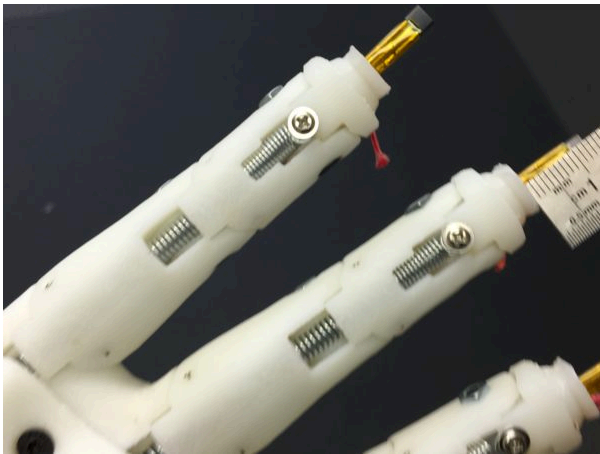
Solder the second and third wire and wrap them up with tape



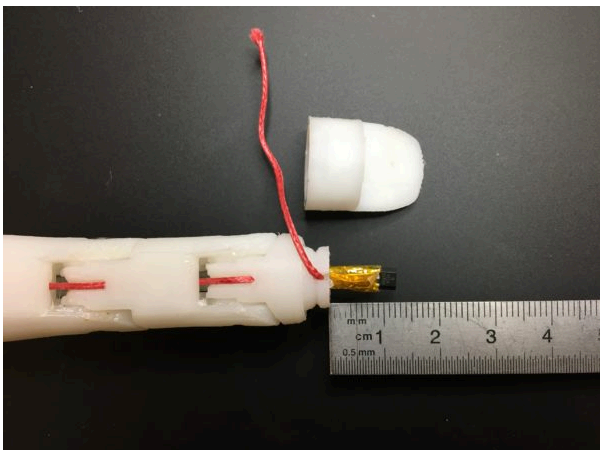
(<https://inmoov.fr/>)



When wrapping the tape, you shouldn't make it thicker than the ribbon itself, otherwise you will have trouble pushing it down for the next step.



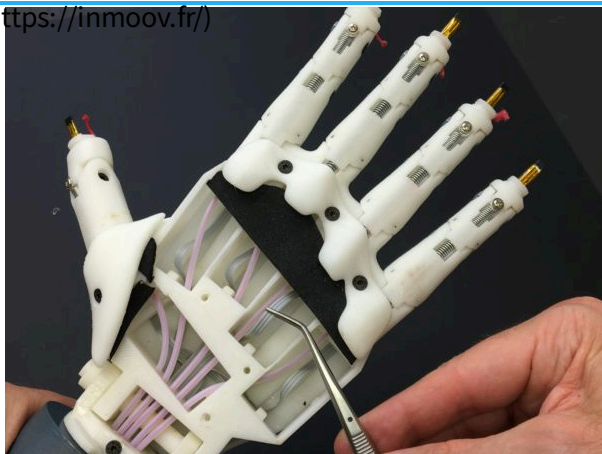
Push it back into the finger tip carefully



This is the length of what should be sticking out of the



(<https://inmoov.fr/>)



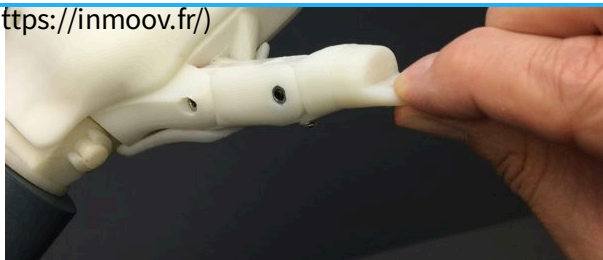
You can pull the wires in the palm to help yourself to push the hall sensor back into the tip



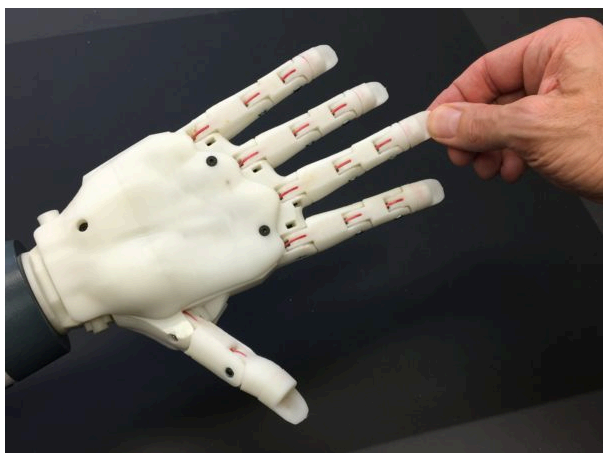
Wrap the fishing line around the groove, and carefully screw the end of the finger over the sensor.



(<https://inmoov.fr/>)



Note the end of the finger is printed with a small threading to allow it to be screwed clockwise to the finger tip. Don't force, otherwise the threading will be damaged.



Finish the operation for each finger.

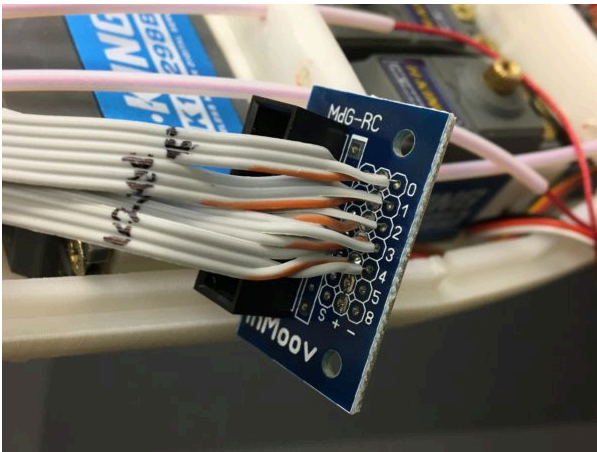




(<https://inmoov.fr/>)



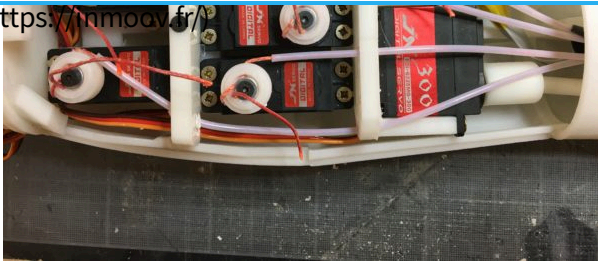
Time to solder each wires to the sensor mini board of the NervoBoard. Thumb goes on « 0 », Index on « 1 », Majeure on « 2 », RingFinger on « 3 », Pinky on « 4 »



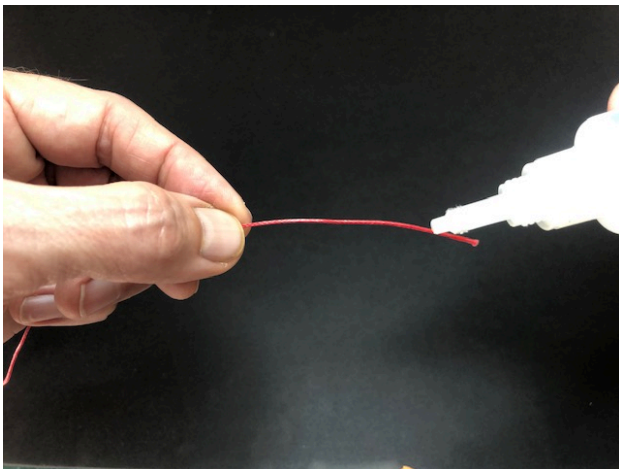
You can see more info about the sensor connection on this page: <https://inmoov.fr/test-your-finger-sensor/> (<https://inmoov.fr/test-your-finger-sensor/>)



(<https://inmoov.fr/>)



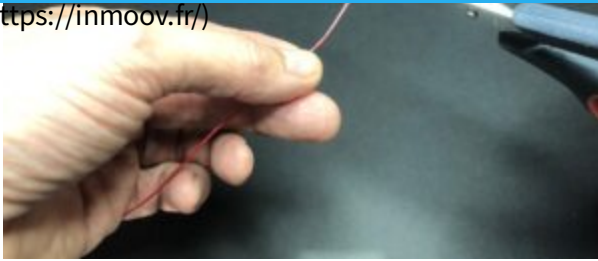
Here you can see where I have set my mini analog board. You can also see on the pulleys, how the screws with a small washer are holding the fishing line in place with a single knot. To do that, you need to add some glue on your fishing braided lines at the point you want to cut them.



To do that, you need to add some glue on your fishing braided lines at the point you want to cut them, let the contact glue cure. The glue makes it easier to get a clean cut.



(<https://inmoov.fr/>)



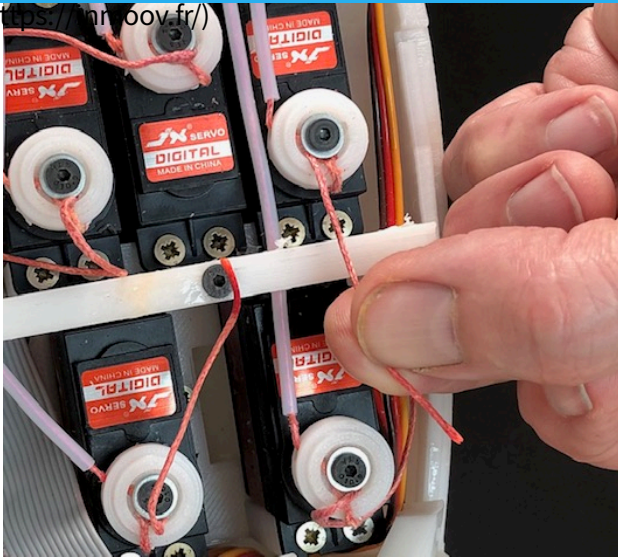
Then, use sharp scissors to cut where you have put glue.



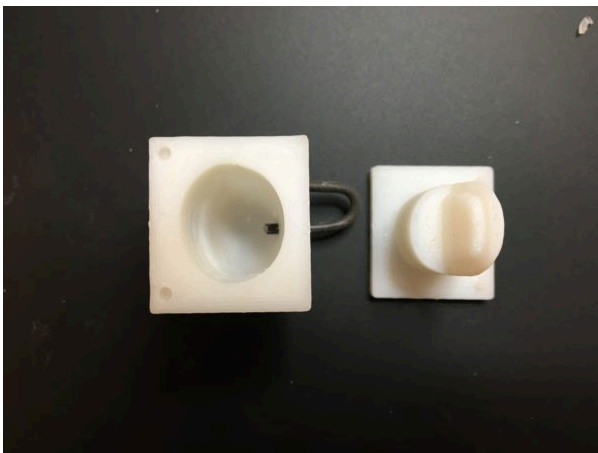
Run your fishing line the way I did it in this picture, it should make a big knot.



(<https://inmoov.fr/>)



Tight the knot, set your screw and washer on the knot, and maintain the knot pulled while finishing tightening the screw. .



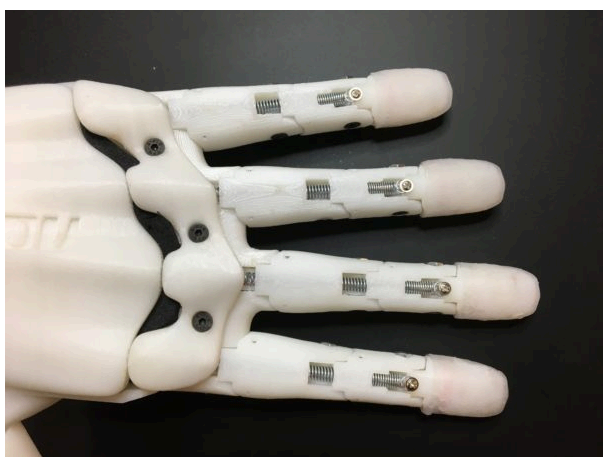
*In the molds, there is a hole to insert a section of 2mm diameter, which will allow to create a reset in the molded silicone for the magnet to take place. Disc Magnet diameter 2.5mmx1mm
 (https://www.supermagnete.fr/eng/disc-magnets-neodymium/disc-magnet-2.5mm-1mm_S-2.5-01-N52N)*



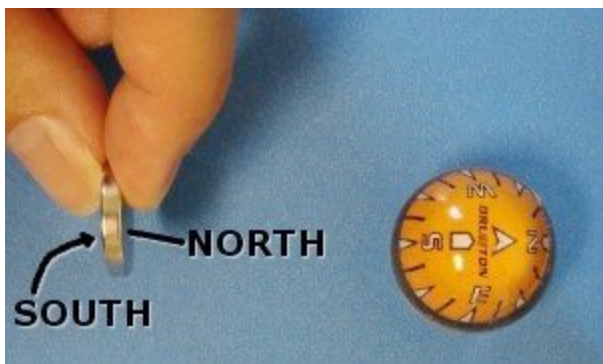
(<https://inmoov.fr/>)



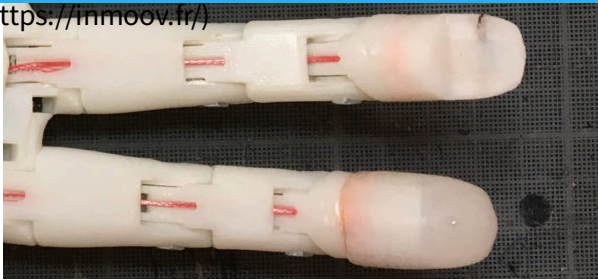
Make it go through of 2mm. I used two component clear RTV silicone Ecoflex™ 00-10



Glue with silicone kit the molded parts on each finger tip



Determine the South pole of your magnet with a compass.


[\(https://inmoov.fr/\)](https://inmoov.fr/)


Insert the magnet with the South pole facing the hall sensor into the reset of the silicone molded tip. Use wooden toothpick to do that, because any metal tool will stick to the magnet. This is delicate to do, and should be well done to get proper results. Secure the magnet position with a drop of silicone kit, and wait for it to dry. Test your sensors with this page <https://inmoov.fr/test-your-finger-sensor/> (<https://inmoov.fr/test-your-finger-sensor/>)

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